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Trends in the Use of Interactive Digital Comics in Indonesian Language Learning for Lower-Grade Elementary Schools: A Systematic Review 2020–2025

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Abstract
Reading literacy among Indonesian elementary school students remains relatively low, as reflected in Indonesia's PISA 2022 reading score of 359, which is considerably below the OECD average. This condition underscores the urgent need for innovative, technology-based learning media, including interactive digital comics. Despite the increasing number of studies on digital learning media, systematic evidence specifically addressing interactive digital comics in Indonesian language learning for lower-grade elementary students remains limited. This study aimed to identify research trends, examine learning effectiveness, and map the strengths, limitations, and research gaps related to interactive digital comics published between 2020 and 2025. This study employed a Systematic Literature Review (SLR) using the PRISMA 2020 framework. Literature was collected from Google Scholar, Scopus, and ERIC databases. From 800 identified articles, 30 studies met the inclusion criteria after screening, eligibility assessment, and quality evaluation using the Mixed Methods Appraisal Tool (MMAT). Data were analyzed through thematic and narrative synthesis, supported by bibliometric mapping using VOSviewer. The findings revealed a significant increase in studies on interactive digital comics after 2021, particularly on Android- and web-based platforms integrated with multimedia features. Most studies reported positive effects on reading comprehension, vocabulary mastery, learning motivation, and student engagement. However, research gaps remain in longitudinal studies, large-scale experiments, local cultural integration, and the assessment of affective learning. Interactive digital comics demonstrate strong potential as multimodal learning media for elementary language education.

INTRODUCTION

Reading proficiency constitutes the primary cognitive foundation for academic success across all subject areas, and its mastery is especially critical during the early years of formal schooling (Harpiani, 2021; Tarigan & G, 2015). At the elementary level, particularly in lower grades, the acquisition of reading comprehension, vocabulary, and decoding skills determines students' capacity to engage meaningfully with Indonesian language instruction in subsequent grade levels (Serafini, 2020). These foundational competencies are not merely linguistic in nature; they form the basis of cognitive engagement with all academic content, making early literacy development one of the most consequential educational priorities in any national schooling system (Kirby & Savage, 2008; Mol & Bus, 2011). Despite the centrality of reading literacy to educational outcomes, empirical evidence

consistently indicates that Indonesian students' achievement in this domain remains considerably below international benchmarks.

The 2022 Programme for International Student Assessment (PISA), released by the OECD in December 2023, recorded Indonesia's reading literacy score at 359 points, a 12-point decline from the 2018 figure of 371, marking the lowest score since Indonesia first participated in PISA in 2000, and sitting far below the OECD average of 476 points (OECD, 2023). Indonesia ranked 69th out of 80 participating countries, with more than 60% of students scoring at Level 1 or below in reading (OECD, 2023). Longitudinal analyses confirm that this downward trend is not isolated; students who enter the upper elementary grades as weak readers continue to struggle throughout their schooling, indicating that the problem is deeply rooted in the foundational stages of literacy acquisition (Francis et al., as cited in Kim et al., 2021; Binks-Cantrell et al., 2020). At the national level, the 2023 National Assessment data showed that only 61.53% of elementary school students met literacy competencies above the minimum threshold, an increase from 53.42% in 2021, but still leaving more than 38% below the minimum standard. The 2022 Public Literacy Index (ILM) survey further revealed that Indonesian children's text comprehension stood at only around 15%, while the average comprehension of elementary students toward informational texts hovered at roughly 33% (Muliawanti et al., 2022). These figures collectively signal a systemic literacy crisis that warrants urgent pedagogical intervention.

This literacy gap does not arise in isolation. Research consistently demonstrates that lower-grade learners experience systematic difficulties across the four foundational language skills, listening, speaking, reading, and writing (Ubaidillah et al., 2025; Damopolii et al., 2021), difficulties that are often compounded by instructional approaches that fail to align with the developmental characteristics of young children. Cognitively, students in Grades I–III are transitioning from the preoperational to the concrete operational stage of development in Piaget's framework, a phase during which logical reasoning remains tethered to tangible, perceptible referents rather than abstract representations (Piaget, 1964; Santrock, 2011). In this stage, visual representation serves as an essential cognitive scaffold, bridging the gap between abstract linguistic content and concrete understanding (Hakim & Ahmadi, 2025; Serafini, 2020). When learning materials are monotonous, text-heavy, and visually unresponsive to this developmental reality, they systematically fail to engage young learners (Ahmad & Mustika, 2021; Reinita et al., 2023). The resulting disengagement has long-term consequences: inadequate literacy foundations acquired in the lower grades predictably hinder students' ability to meet the demands of higher-level academic learning (Binks-Cantrell et al., 2020).

Against this backdrop, interactive digital comics emerge as a theoretically grounded and empirically promising alternative. As a medium that inherently integrates narrative text, visual imagery, audio, animation, and interactive elements, interactive digital comics align closely with the principles of Dual Coding Theory (DCT) proposed by Paivio (1986, 2007), which holds that information comprehension and retention increase significantly when verbal (linguistic) and nonverbal (visual imagery) systems are activated simultaneously (Clark & Paivio, 1991; Li et al., 2022). Mayer (2009, 2021) extended this theoretical framework through the Cognitive Theory of Multimedia Learning (CTML), demonstrating that meaningful learning occurs when verbal and visual information are presented in a structured and coherent manner, enabling learners to construct integrated mental representations. Empirical research grounded in these frameworks has consistently shown that combining visual and textual modalities improves reading retention, vocabulary acquisition, and comprehension in young learners (Zhang et al., 2026; Castillo-Cuesta & Quinonez-Beltran, 2022; Freire Cabrera & Perea Beltrán, 2024). Interactive digital comics transcend their conventional counterparts by incorporating additional interactivity features, such as responsive quizzes, narrative audio, branching storylines, and student-paced navigation, which operationalize the CTML principles of segmentation and learner control (Mayer, 2021; Rasch & Schnotz, 2009). Interactivity grants students agency over the pace and sequence of information processing, thereby theoretically reducing extrinsic cognitive load while enhancing generative processing (Mayer & Moreno, 2003; Lin et al., 2021).

Nevertheless, when interactivity is designed without a coherent theoretical framework, it risks generating cognitive overload in children whose working memory capacity is not yet fully mature, a risk underscored by multiple studies on multimedia design for young learners (Mayer & Moreno, 2003; Sweller, 1994). The specific conditions under which interactivity enhances rather than impedes learning in primary educational contexts thus constitute a critical, insufficiently answered empirical question.

Despite a growing body of literature on digital learning media, a rigorous synthesis of evidence specifically addressing interactive digital comics in Indonesian language learning for lower-grade elementary students remains largely absent. Prior reviews have addressed digital media use in general or higher education contexts (Maulidia et al., 2025), evaluated the sustainability of media effects across subjects without specific focus on early literacy (Janah & Aeni, 2025), and identified the need for further inquiry into interactive comics for elementary learners without providing systematic synthesis (Alifah et al., 2025). Individual studies have similarly demonstrated effectiveness in isolated contexts, such as R&D studies producing valid and practical media (Handayani et al., 2023); Khairunnisak et al., 2024) and experimental studies confirming improvements in reading comprehension (Wijaya et al., 2025; Wang et al., 2026), yet without systematic cross-study synthesis, it remains impossible to draw reliable conclusions about trends, cumulative effectiveness, design characteristics, and research gaps across the field. Moreover, no prior systematic review has specifically mapped the development trends, multimedia characteristics, and knowledge gaps pertaining exclusively to interactive digital comics in Indonesian language learning for lower-grade primary students within the 2020–2025 period. This represents a substantive gap in the existing evidence base.

This study addresses that gap by conducting a Systematic Literature Review (SLR) using the PRISMA 2020 framework to synthesize empirical studies published between 2020 and 2025. Unlike previous studies, this review specifically focuses on interactive digital comics, as distinct from static digital or print comics, used in Indonesian language instruction for Grades I–III students, and it integrates evidence on development trends, effectiveness dimensions, multimedia characteristics, and unresolved research gaps. The study establishes three principal objectives: (1) to identify trends in the development of interactive digital comics in elementary school Indonesian language learning between 2020 and 2025; (2) to analyze their effectiveness on learning outcomes among lower-grade students, particularly in the dimensions of reading comprehension, vocabulary mastery, and learning motivation; and (3) to map the existing strengths, weaknesses, and research gaps in the literature. These objectives respond to the following research questions: (a) What are the trends in the development of interactive digital comics for elementary Indonesian language learning from 2020 to 2025? (b) What is their impact on lower-grade students' learning outcomes? (c) What are the strengths, weaknesses, and research gaps identified in these studies? The findings are expected to provide an evidence-based foundation for educators, media developers, and policymakers seeking to harness the potential of interactive digital comics to address Indonesia's persistent early literacy crisis and contribute to the attainment of equitable, quality education in accordance with SDG 4.

METHODS

This study employed a Systematic Literature Review (SLR) design, a methodological approach selected for its capacity to produce evidence-based, comprehensive, measurable, and reproducible conclusions by systematically identifying, evaluating, and synthesizing findings from a defined body of relevant literature (Kitchenham et al., 2009; Page et al., 2021). The research procedure adhered strictly to the PRISMA 2020 guidelines (Preferred Reporting Items for Systematic Reviews and Meta-Analyses), which organizes the review process into four sequential phases: identification, screening, eligibility assessment, and inclusion. This framework was adopted to ensure transparency, rigor, and methodological consistency throughout the synthesis process, and its application is aligned with the three research objectives of this study, identifying development trends, analyzing learning

effectiveness, and mapping research gaps in the use of interactive digital comics in Indonesian language learning for lower-grade elementary students.

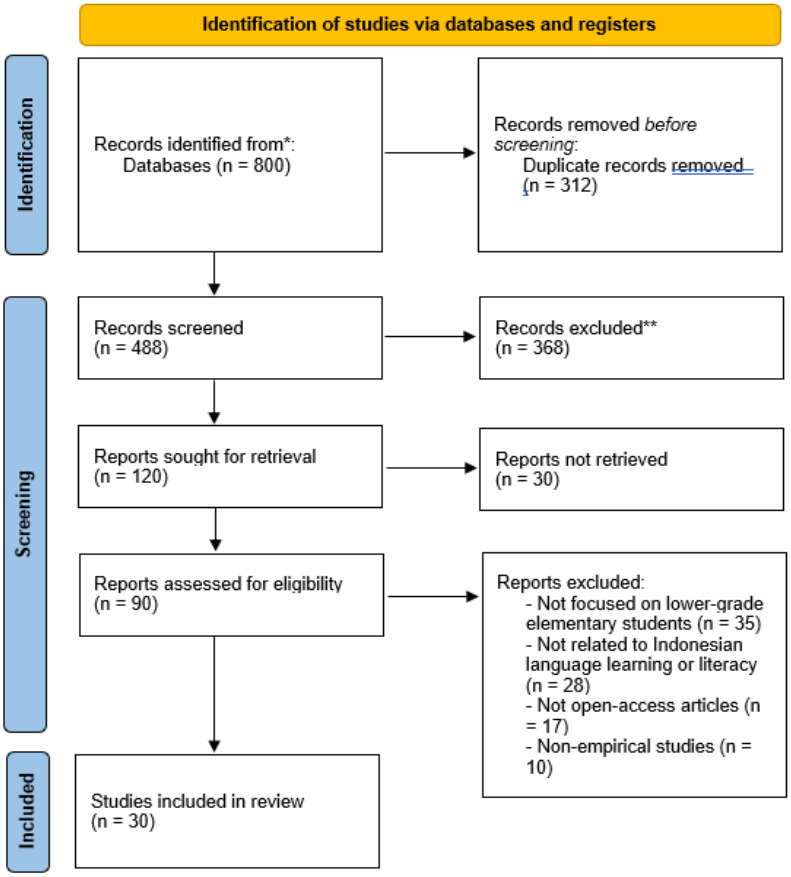


Figure 1. PRISMA Flow Diagram

The literature search was conducted across three electronic databases: Google Scholar, selected to capture open-access Indonesian-language publications; Scopus, for internationally peer-reviewed articles; and ERIC (Education Resources Information Center), for education-focused literature. These searches yielded 410, 251, and 139 articles respectively, producing an initial pool of 800 records. To maximize retrieval coverage, a combination of Indonesian and English keywords was applied, including: komik digital pembelajaran, komik interaktif sekolah dasar, digital comics learning elementary school, interactive comics language learning, comic-based learning Indonesian language, and media komik digital hasil belajar. Articles were subsequently screened against a set of pre-established inclusion and exclusion criteria. Inclusion criteria required that studies: (a) were empirical articles published between 2020 and 2025; (b) examined the use of digital or interactive comics with lower-grade elementary school students (Grades I–III) as the target population; (c) employed empirical research designs, including experiments, research and development (R&D), or quantitative/qualitative methods; and (d) were available in full text under open access in Indonesian or English. Articles were excluded if they were non-empirical, irrelevant to Indonesian language learning or reading literacy, did not specifically address interactive digital comics, targeted populations outside lower-grade elementary students, or lacked open-access full-text availability. It is noted that several articles, including Freire Cabrera & Perea Beltrán (2024) and Mayer (2021), were retained as theoretical and comparative benchmarks owing to their substantive cross-contextual relevance; their inclusion is transparently documented in the synthesis table.

The article selection process was conducted independently by two researchers at each stage to minimize selection bias. Inter-rater reliability was established using Cohen's Kappa coefficient,

which produced a value of κ = 0.82, classified as indicating strong agreement (Landis & Koch, 1977). Any discrepancies between reviewers were resolved through structured consensus discussions. Following initial duplicate removal, which eliminated 312 records, the remaining 488 articles underwent title and abstract screening, resulting in the exclusion of 368 articles that did not address interactive digital comics in Indonesian educational contexts. Of the 120 articles subjected to full-text eligibility review, 90 were excluded for the following reasons: 35 did not involve lower-grade elementary students, 28 lacked an Indonesian language or literacy learning focus, 17 were not open access, and 10 were non-empirical. This sequential screening process yielded a final corpus of 30 articles, the flow of which is documented in the attached PRISMA 2020 diagram (Figure 1). Methodological quality of the included studies was assessed using an adapted version of the Mixed Methods Appraisal Tool (MMAT), which evaluated each article across five dimensions: clarity of research objectives, appropriateness of design, adequacy of data collection procedures, validity of instruments, and rigor of data analysis. Based on this evaluation, 11 articles (36.70%) were classified as high quality, 16 (53.30%) as medium quality, and 3 (10.00%) as low quality (see Table 1).

Data were extracted using a structured extraction sheet that captured the following information from each included study: author names, publication year, research design, target grade level, digital comic platform type, embedded interactive features, Indonesian language instructional material covered, measured learning outcomes, principal findings, and reported study limitations. Extracted data were then analyzed through two complementary approaches. Thematic synthesis was employed to identify convergent patterns across studies regarding media characteristics, instructional effectiveness, and temporal research trends, while narrative synthesis was used to qualitatively interpret relationships among study outcomes and contextualize findings within the theoretical frameworks of Dual Coding Theory and the Cognitive Theory of Multimedia Learning. In addition, a bibliometric analysis was conducted using VOSviewer (version 1.6.20) to map keyword co-occurrence networks and trace the evolution of research focus within the field between 2020 and 2025.

RESULTS AND DISCUSSION

Results

General Characteristics of Analyzed Articles

In accordance with the PRISMA 2020 protocol, 30 articles met the full inclusion criteria and were selected for final synthesis. Table 3 presents the distribution of these articles by year of publication, revealing a clear upward trajectory in research output over the review period. Only two articles were published in 2020, with three in 2021. Output then accelerated markedly, rising to seven articles in 2022 and reaching a peak of nine articles in 2023. The subsequent year registered six articles in 2024, followed by three articles from early 2025. This pattern indicates a sharp and sustained escalation in academic attention to interactive digital comics in elementary education following 2020.

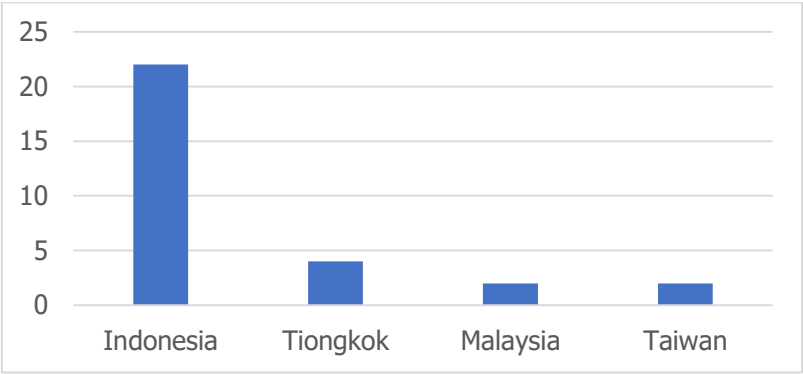


Figure 2. Country of Origin Publication Graph

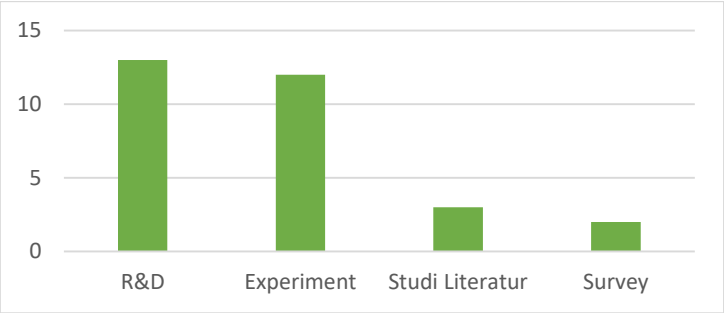


Figure 3. Research Design Graph

Geographically, Indonesian researchers dominated the corpus, accounting for 22 of the 30 included studies (73.3%). The remaining eight studies originated from China (n = 4), Malaysia (n = 2), and Taiwan (n = 2). In terms of research design, Research and Development (R&D) was the most frequently applied methodology (n = 13, 43.3%), followed closely by experimental designs (n = 12, 40.0%). The remaining five articles employed literature reviews (n = 3) or survey approaches (n = 2), reflecting the field's predominant orientation toward media development and empirical testing rather than large-scale measurement or population-level inquiry.

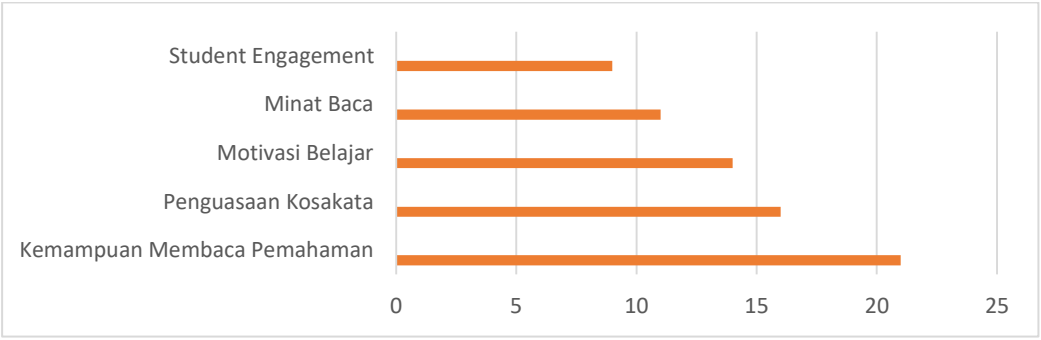


Figure 4. Graph of measured variables

The research subjects were predominantly drawn from Grades II and III. As shown in Figure 4, the five most frequently measured learning outcome variables across the 30 studies were reading comprehension (n = 21, 70.0%), vocabulary mastery (n = 16, 53.3%), learning motivation (n = 14, 46.7%), reading interest (n = 11, 36.7%), and student engagement (n = 9, 30.0%). Notably, no study within the corpus explicitly measured psychomotor outcomes, and only three studies reported affective learning outcomes beyond motivation, an unexpected finding that underscores a systematic under-representation of holistic developmental dimensions in the extant literature.

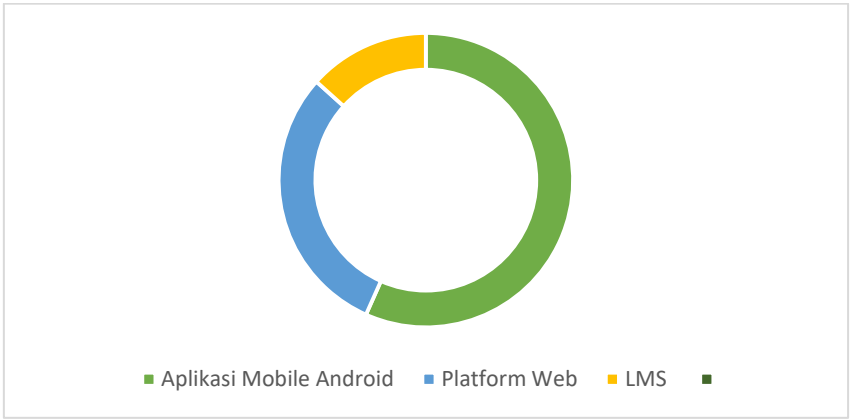


Figure 5. Interactive digital media usage platform

With respect to platform deployment, Android-based applications were the most prevalent, accounting for 56.7% of the media analyzed, followed by web-based platforms (30.0%) and Learning Management System (LMS)-integrated formats (13.3%). The majority of digital comic products in the corpus integrated at least two interactive features simultaneously, the most common combinations being narrative audio with animated visual elements (n = 17), multiple-choice embedded quizzes (n = 14), and gamified reward systems such as scoring mechanisms and progress badges (n = 11).

Characteristics of Effective Interactive Digital Comics

Synthesis of data across the 30 included articles revealed four primary design characteristics that were consistently and convergently associated with instructional effectiveness. First, in terms of narrative structure, studies reporting the highest learning gains uniformly adopted a linear story arc, specifically the orientation–complication–resolution sequence, tailored to the cognitive developmental level of lower-grade students and grounded in daily life contexts (Ayudianingsih et al., 2025; Pramesti et al., 2024). Second, with respect to visual design, studies that documented measurable improvements in reading comprehension and motivation consistently featured high-contrast, brightly colored illustrations; expressive character designs; and typographically legible text in adequate font sizes. Third, concerning interactivity, the most effective media integrated layered interactive elements, including "choose your own story" branching paths, drag-and-drop vocabulary activities, and immediate corrective feedback following embedded quiz responses. Fourth, with regard to curriculum alignment, 25 of the 30 analyzed media products were explicitly designed to align with Indonesian curriculum competency standards, with the majority targeting reading comprehension, vocabulary enrichment, and narrative text comprehension as their primary learning objectives.

Research Article Synthesis

The overall synthesis of 30 included articles confirmed that interactive digital comics exerted a consistently positive influence on learning outcomes in Indonesian language education at the elementary school level. Specifically, 25 of the 30 studies (83.3%) documented measurable improvements in at least one learning outcome dimension (reading comprehension, vocabulary mastery, or learning motivation) compared to conventional instructional approaches. The remaining five studies (16.7%), primarily those using survey or literature review designs, did not include direct pre-post outcome measurements and therefore reported potential effectiveness rather than empirically confirmed results. Table 1 presents a comprehensive matrix of all analyzed articles, organized to illustrate the distribution of research designs, instruments, key findings, and reported limitations across the 2020–2025 period.

Table 1. Synthesis of Interactive Digital Comics in Indonesian Language/Literacy Learning (2020–2025)

No	Article Title	Author & Year	Method	Instrument	Key Findings	Limitations
1	Revitalisasi Pembelajaran Bahasa Indonesia melalui Komik Digital untuk Peningkatan Kemampuan Membaca di Sekolah Dasar	Ayudianingsih et al. (2025)	R&D (ADDIE)	Expert validation, pretest-posttest, N-Gain	Canva-based digital comics improve Grade II reading ability; mean scores rose from 55% pre-intervention; media rated valid and practical	Limited to a single school; no control group
2	Komik Digital Berbasis Kearifan Lokal untuk Meningkatkan Literasi Membaca Siswa Sekolah Dasar	Mekalungi et al. (2025)	Quantitative experiment	Reading literacy test	Local-wisdom-based digital comics significantly improve reading literacy scores in elementary students	Sample restricted to one regency; no longitudinal tracking
3	The Use of Comics to Develop EFL Students´ Reading Comprehension and Vocabulary	Freire Cabrera & Perea Beltrán (2024)	Experiment (quasi-experimental)	Vocabulary test, reading comprehension test	Digital comics significantly improve vocabulary scores and reading	Sample limited to one geographic region;

	Acquisition				comprehension compared to print-based media	Mandarin-language context
4	Komik Digital Berbasis Kontekstual untuk Meningkatkan Hasil Belajar Bahasa Indonesia Siswa Kelas II Sekolah Dasar	Pramesti et al. (2024)	R&D (ADDIE)	Expert validation, learning outcomes test	Contextual digital comics improve Grade II Indonesian language learning outcomes; media rated valid and practical	Trial limited to one school; no control group
5	Pengembangan Bahan Ajar Komik Digital Berbasis Web Pixton Pada Mata Pelajaran Bahasa Indonesia Fase C Sekolah Dasar	Handayani et al (2023)	R&D (4D)	Expert validation, learning outcomes test	Developed media rated valid, practical, and effective for Indonesian language instruction	Not tested at larger scale; single-school trial
6	Interactive Digital Comic Teaching Materials to Increase Student Engagement and Learning Outcomes	Khotimah & Hidayat (2022)	R&D (ADDIE)	Student engagement questionnaire, learning outcome test	Interactive digital comics proven effective in increasing student involvement and learning outcomes in online independent learning	Context-specific to online COVID-era learning; not early literacy-focused
7	Effects of Interactive Digital Comics on Student Engagement and Learning Achievement	Lin et al. (2021)	Experiment	Learning outcome test, engagement questionnaire	Interactive digital comics significantly improve both engagement and learning achievement compared to conventional media	Not specifically focused on early elementary literacy
8	Penggunaan Komik Digital sebagai Media Pembelajaran untuk Meningkatkan Kemandirian Belajar dan Berpikir Kritis	Sutomo & Kusmaryono (2025)	Experiment	Learning independence scale, critical thinking test	Digital comic media enhances learning independence and critical thinking in elementary students	Affective dimensions not fully measured; limited sample
9	Digital Storytelling Comics for Language Learning: A Multimedia Learning Approach	Zhang et al. (2026)	Experiment	Retention test, transfer test	Visual-verbal combination in digital storytelling comics significantly improves language retention and transfer	Focused on retention; not contextualized to early literacy in Indonesian elementary schools
10	Educational Digital Comics to Improve Reading Comprehension Skills Among Fourth-Grade Elementary School Students	Wijaya et al. (2025)	R&D (ADDIE)	Reading comprehension test	Android-based digital comics produce significantly higher reading comprehension scores than conventional text	Relatively simple experimental design; no effect size reported
11	Penerapan Media Komik Digital untuk Meningkatkan Keterampilan Membaca Nyaring Peserta Didik Kelas I	Ernayanti & Winarni (2024)	Classroom Action Research (CAR)	Oral reading assessment, observation	Digital comics effectively improve Grade I oral reading skills across two cycles; classical completion reached 100% in Cycle II	Small sample (10 students); single school; no control condition
12	Digital Comics for Improving EFL Vocabulary Learning in Secondary School	Castillo-Cuesta & Quinonez-Beltran (2022)	Experiment	Vocabulary assessment	Digital comics significantly improve vocabulary acquisition compared to traditional methods; students report higher motivation	Secondary school context; EFL rather than Indonesian L1 setting
13	Studi Kesiapan Guru Dalam Mengintegrasikan Peran Teknologi Digital pada Pembelajaran Kurikulum Merdeka di Sekolah Dasar	Afnani & Attalina (2025)	Survey	Teacher digital competency questionnaire	Teacher digital competency varies considerably; limited technical skills constitute a primary barrier to effective digital media use	No intervention conducted; descriptive only

14	Komik Digital Berbasis Kearifan Lokal Melayu Riau untuk Materi Persatuan dan Kesatuan	Agustin & Ramadan (2025)	R&D (ADDIE)	Material expert validation, media expert validation, student response questionnaire	Local-wisdom-integrated digital comic media rated valid and practical for Grade V	No effectiveness test conducted; no pre-post measurement
15	Conducting concept map-based generative AI-assisted comic storytelling activities in EFL environments: performance differences among students with different AI literacy levels	Wang et al. (2023)	Experiment	Reading comprehension test	Experimental group achieves significantly higher reading scores than control group; visual-textual integration supports comprehension	Cultural contexts not explored; Chinese-language school setting
16	Pengembangan Komik Digital Berbasis CTL untuk Pemahaman Konsep dan Motivasi Belajar Siswa SD	Rahayu et al. (2023)	R&D	Expert validation, motivation scale, comprehension test	CTL-based digital comics improve concept comprehension and learning motivation among elementary students	Limited to science content; small sample; single school
17	Pengaruh Media Komik Digital untuk Menumbuhkan Minat Baca Siswa Kelas 4 SDN	Fikri et al. (2025)	Experiment	Reading interest scale (Likert), Paired Sample T-test	Reading interest scores increase from 52% (pre) to 84% (post); statistically significant (p = .000)	Grade 4 sample only; no longitudinal follow-up
18	Digital Comics in Online Learning During COVID-19: Effects on Student Cognitive Learning Outcomes	Damopolii et al. (2021)	Experiment	Cognitive learning outcomes test	Digital comics in online learning positively affect cognitive outcomes; significantly higher scores in digital comic group	COVID-19 context limits generalizability to face-to-face settings
19	Pengembangan E-Komik Berbasis Literasi Numerasi di Kelas IV Sekolah Dasar	Afifah et al. (2024)	R&D	Expert validation, student response	Numeracy-literacy e-comic rated valid and engaging; students respond positively to format	Numeracy-focused; does not directly measure Indonesian language literacy outcomes
20	E-Comic-Based Interactive Learning Media in Elementary Grades	Khairunnisak et al. (2024)	R&D	Expert validation, learning outcomes test	Interactive e-comic rated highly suitable for lower elementary grade use; positive expert validation	Relatively simple trial design; limited theoretical integration
21	Feasibility of Digital Comic Media in Enhancing Early Reading Skills for Grade I Elementary School Students	Ananda et al. (2024)	R&D	Expert validation, early reading skill test	Digital comic media demonstrated high feasibility for improving Grade I early reading skills	Single-school trial; no control group; no long-term evaluation
22	Using Digital Comics for Enhancing EFL Vocabulary Learning during COVID-19 Pandemic	Castillo-Cuesta (2022)	Experiment	Vocabulary assessment, motivation scale	Digital comics significantly enhance EFL vocabulary learning; motivational benefits observed	COVID-19 online context; secondary-level EFL context
23	Pengembangan Media Komik Digital untuk Pembelajaran PPKn di SD/MI	Antika et al. (2025)	R&D (literature review)	15 relevant publications analyzed	R&D model dominates development; media rated valid and effective across reviewed studies	Literature review methodology; no original primary data
24	Multimedia Learning (3rd ed.)	Mayer (2021)	Theoretical benchmark	Retention test (cited experiments)	Visual+verbal integration effectively improves learning retention; CTML principles empirically supported	General multimedia context; does not specifically measure early childhood

						literacy in Indonesian settings
25	Pengembangan Komik Digital Penerapan Sila Pancasila untuk SD	Akbar et al. (2023)	R&D (ADDIE)	Expert validation, questionnaire, learning outcomes test	Media is valid and improves students' understanding of Pancasila principles	Limited sample; Civics-focused rather than language-focused
26	Komik Digital Berbasis Kearifan Lokal untuk Meningkatkan Literasi Membaca	Wulandari et al. (2023)	Survey/needs analysis	Document analysis, teacher & student questionnaire	Digital comics needed to improve reading literacy; local wisdom integration strongly desired by stakeholders	Needs analysis only; no intervention data
27	Incorporating Multimodal Literacies into Classroom-Based Reading Assessment	Serafini (2020)	Literature study	Visual text analysis; multimodal assessment framework	Visual and multimodal resources significantly support young readers' comprehension of contemporary texts	Theoretical/framework paper; not empirical experiment in Indonesian context
28	Developing Digital Comics as Character Learning Media for Elementary School Students	Rohmanurmeta & Dewi (2020)	R&D	Expert validation, student response questionnaire	Digital comics rated valid as character education media; students respond positively	Character-focused; no direct measurement of language competency outcomes
29	Pengaruh Media Komik Digital terhadap Minat dan Hasil Belajar Siswa pada Pelajaran IPA	Rahmah et al. (2025)	Experiment	Reading interest scale, science learning outcomes test	Digital comic media significantly improves learning interest and science outcomes in elementary students	Science-subject focus; affective and psychomotor dimensions underexplored
30	Interactive Digital Comic Learning Media Based on Literacy in Social Studies for Elementary School Students	Wijayanti et al. (2025)	R&D	Expert validation, student questionnaire	Literacy-based interactive digital comic media rated valid, practical, and engaging for elementary Social Studies	Social Studies content focus; no longitudinal effectiveness measurement

Bibliometric Analysis

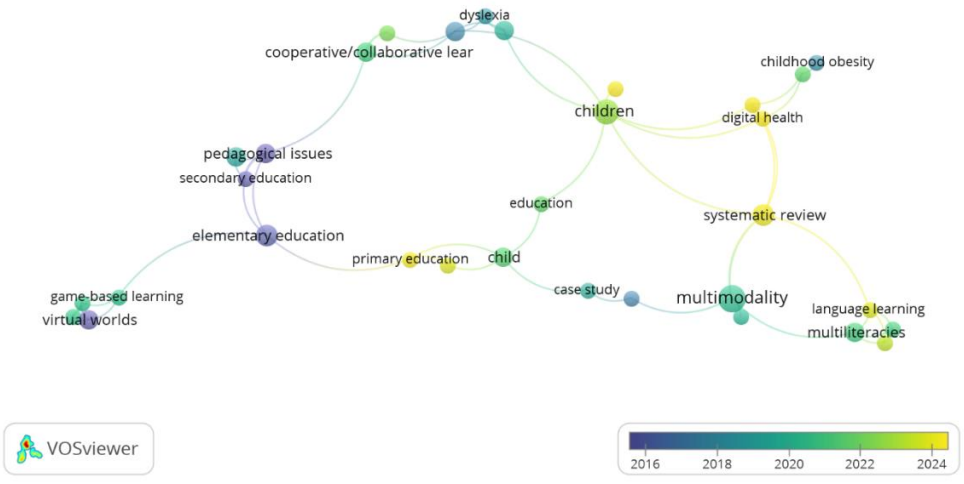


Figure 6. VOSviewer Co-occurrence Keywords Analysis Results

Keyword co-occurrence analysis conducted via VOSviewer (version 1.6.20) generated five core thematic clusters: digital comics, language learning, literacy, multimodality, and elementary education. Chronologically, the bibliometric mapping reveals three distinct developmental phases in the research landscape. During the initial phase (2016–2018), the dominant keywords were elementary education

and pedagogical issues, with isolated references to virtual worlds and early digital tools. The transition phase (2019–2021) witnessed a notable surge in terms such as game-based learning, multimodality, and inclusive education, specifically referencing interventions for students with dyslexia, signaling an initial pivot from general pedagogy toward more differentiated and technology-mediated instructional approaches. The recent phase (2022–2024) is characterized by the pronounced emergence of keywords including language learning, multiliteracies, and systematic review, indicating that the field has entered a consolidation period in which interactive digital comics are recognized as dedicated vehicles for literacy advancement. A noteworthy finding from the bibliometric analysis is the relative absence of terms such as teacher training, infrastructure, and equitable access from the core clusters, despite these emerging as recurring limitations in individual studies, suggesting a disconnect between reported implementation barriers and the theoretical orientation of the field's primary scholarly output.

Discussion

The findings of this systematic review converge to produce three mutually reinforcing conclusions that directly address each of the stated research questions and collectively illuminate the current state of knowledge regarding interactive digital comics in elementary Indonesian language education.

Trends in Research and Platform Development

The bibliometric evidence and temporal publication distribution confirm that academic interest in interactive digital comics as instructional tools in Indonesian elementary education has escalated substantially since 2021, with the peak occurring in 2022–2023. This trajectory closely mirrors the global post-pandemic digital acceleration documented by Suprapto et al. (2024), who conducted a bibliometric analysis of digital comics in education and similarly identified 2021 as the inflection point for accelerated growth in this research subfield. The dominance of Android-based platforms (56.7%) in the analyzed corpus is consistent with patterns reported by Listianingsih et al. (2021) and corroborated internationally by Damopolii et al. (2021), who demonstrated that mobile-based digital comic delivery is particularly effective in contexts where device access is primarily through smartphones rather than desktop computers. The predominantly lower-income household profile of Indonesian elementary school families means that Android-based media represent a strategically pragmatic accessibility choice, a design decision that Reinita et al. (2023) explicitly identified as critical for successful adoption in field settings.

The bibliometric shift toward multiliteracy and multimodal paradigms is theoretically significant. Kress & van Leeuwen's (2001) social semiotic framework anticipated precisely this transition, arguing that meaning-making in the twenty-first century is inherently multimodal rather than exclusively linguistic. The evolution from passive digital comics to interactive multimodal environments directly operationalizes New Literacies perspectives articulated by Lankshear & Knobel (2011), wherein digital texts are understood as participatory, networked, and non-linear. The present review adds an empirical dimension to these theoretical claims by demonstrating that this theoretical shift has been accompanied by a measurable increase in experimental evidence confirming effectiveness, a pattern not yet synthesized in previous literature.

An unexpected finding within the trend analysis is the relatively limited representation of studies from regions outside Java (only 4 of 22 Indonesian studies sampled from outer islands), despite the well-documented urban-rural and inter-regional literacy disparities in Indonesia (OECD, 2023). This geographic concentration suggests that the research base, while expanding, has not yet adequately captured the realities of under-resourced school contexts, a gap that significantly limits the external validity of current effectiveness claims.

Effectiveness of Interactive Digital Comics on Learning Outcomes

The finding that 83.3% of included studies reported positive effects on at least one learning outcome dimension is consistent with, and extends, the broader international evidence base on

multimedia-enhanced language learning. This aggregate effectiveness figure aligns with the conclusions of Freire Cabrera & Perea Beltrán (2024), who demonstrated significant vocabulary and reading comprehension gains using digital comics in a controlled experiment, and with Zhang et al. (2026), whose digital storytelling study confirmed that visual-verbal combinations produce measurably superior language retention and transfer compared to text-only conditions. The theoretical explanation for these convergent findings is provided by Dual Coding Theory (Paivio, 1986, 2007), which predicts that simultaneous activation of the verbal and nonverbal cognitive systems creates redundant representational pathways that strengthen encoding and facilitate retrieval, a mechanism empirically documented in the multimodal vocabulary learning literature by Li et al. (2022) using a mixed-method approach. Mayer's (2009, 2021) Cognitive Theory of Multimedia Learning offers a complementary process-level account, specifying through the coherence, signaling, and temporal contiguity principles that well-structured visual-verbal integration reduces extraneous cognitive load and promotes generative processing, the mechanism through which meaningful, transferable learning is produced.

In the domain of reading comprehension, the present findings closely parallel those of Serafini (2020), who demonstrated that multimodal texts substantially support young readers' meaning-making by providing multiple semiotic entry points into content, and corroborate the classroom-level evidence reported by Pramesti et al. (2024) and Ayudianingsih et al. (2025), whose R&D studies documented significant pretest-to-posttest gains in Grade II Indonesian language comprehension following digital comic interventions. For vocabulary acquisition, the visual-narrative embedding mechanism identified across the reviewed studies, in which target vocabulary is contextualized within emotionally engaging story contexts rather than presented in decontextualized word lists, mirrors the word-learning advantages of contextual embedding documented in the developmental psycholinguistics literature (Beck et al., 2013). With respect to learning motivation, the consistent finding that expressive visual design and interactive features significantly elevate student engagement is supported by Self-Determination Theory (Ryan & Deci, 2020), which identifies perceived competence, autonomy, and relatedness as the three core drivers of intrinsic motivation. Interactive digital comics uniquely address all three: quizzes and branching paths provide autonomy; immediate corrective feedback supports perceived competence; and character-driven narrative fosters emotional relatedness to the learning content.

However, a critical qualification is essential. The positive effectiveness findings must be interpreted with appropriate caution given significant methodological heterogeneity across the corpus. A majority of the included studies relied on single-group pre-post designs without randomization or control conditions, adopted small convenience samples from single schools, and reported intervention periods of as little as two to four weeks. The absence of effect size reporting in 22 of the 30 studies further constrains the precision with which cumulative effectiveness can be quantified. Critically, none of the included studies examined effects beyond the immediate post-intervention period, leaving open the fundamental question of whether observed gains are sustained or whether they reflect transient novelty effects, a distinction with direct implications for educational policy. Castillo-Cuesta & Quinonez-Beltran (2022) raised a directly analogous concern in their study of digital comics in EFL vocabulary learning, cautioning that short-term motivational boosts do not necessarily translate to durable learning outcomes, especially in the absence of deliberate spaced-practice and retrieval mechanisms.

Research Gaps, Strengths, and Implementation Challenges

Three primary strengths of interactive digital comics emerge consistently across the reviewed literature: their inherently engaging and contextual narrative format, which supports intrinsic motivation (Ryan & Deci, 2020; Lin et al., 2021); their capacity for flexible, self-paced access across diverse device types (Damopolii et al., 2021); and their adaptability to varied curricular frameworks, including the Kurikulum Merdeka's competency-based learning objectives (Rahayu et al., 2023; Mekalungi et al., 2025). These strengths position interactive digital comics as a genuinely promising platform for addressing Indonesia's documented early literacy deficit (OECD, 2023).

Conversely, three structural implementation barriers were identified with notable consistency across the reviewed corpus. First, uneven digital infrastructure, particularly in rural and outer-island regions, limits the deployability of digital media at scale (Reinita et al., 2023). Second, variability in teachers' digital competency constrains effective pedagogical integration even where devices are available; Afnani & Attalina (2025) documented this variance empirically in their survey of elementary teachers, and Agustin & Ramadan (2025) corroborated the finding in their development study. Third, the interdisciplinary expertise required for high-quality interactive digital comic development, integrating graphic design, software engineering, instructional design, and pedagogical theory, remains inaccessible to the majority of classroom teachers working in isolation (Freire Cabrera & Perea Beltrán, 2024; Fikri et al., 2025).

Four research gaps emerge from this synthesis with particular urgency. The near-complete absence of longitudinal studies represents the most consequential gap: without evidence of sustained learning gains, the policy value of these media remains undemonstrated (Wang et al., 2023). The paucity of large-scale randomized controlled trials similarly prevents confident causal inference about effectiveness (Zhang et al., 2026). The under-investigation of local cultural integration in comic narratives, despite theoretical arguments from Mekalungi et al. (2025) and Rohmanurmeta & Dewi (2020) that culturally responsive media design amplifies both engagement and comprehension, constitutes a substantive innovation opportunity. Finally, the systematic omission of affective and psychomotor outcome measurement from the majority of studies leaves important dimensions of student development unexamined, in tension with the holistic learning objectives of the Kurikulum Merdeka.

Implications, Limitations, and Conceptual Synthesis

The cumulative findings of this review carry clear implications for educational practice, policy, and future research. For teachers, the evidence suggests that interactive digital comics are most effective when integrated into structured literacy routines with deliberate pedagogical scaffolding, rather than deployed as supplementary novelty tools. For media developers, the data underscore the importance of grounding design decisions in CTML principles, specifically coherence, segmentation, and learner control (Mayer, 2021; Rasch & Schnotz, 2009), while embedding culturally relevant and age-appropriate narrative content aligned with curriculum competency standards. For policymakers, equitable digital infrastructure provision and systematic teacher digital literacy development are necessary prerequisites for the field-level adoption that these media's effectiveness warrants.

This review is subject to four limitations that should contextualize the interpretation of its findings. The search was restricted to three databases, introducing potential publication bias against gray literature and non-English, non-Indonesian language publications. High methodological heterogeneity across included studies precluded quantitative meta-analysis, limiting integration to narrative and thematic synthesis. A small number of included studies featured participants beyond the strict lower-grade elementary target group, handled through transparent documentation in the synthesis matrix. Finally, the review period ends in mid-2025, leaving more recent developments unaddressed.

Conceptually, this review makes a distinctive contribution by demonstrating that interactive digital comics, when theoretically grounded in DCT and CTML and empirically evaluated with appropriate rigor, represent a coherent and reproducible instructional approach rather than a collection of isolated media products. The convergence of 83.3% of studies around positive outcomes, across varied designs, platforms, and grade levels, constitutes meaningful evidence that this media category merits sustained and methodologically rigorous investment. The field's next developmental challenge is to move from proof-of-concept to implementation science: designing studies that can answer not only whether interactive digital comics work, but for whom, under what conditions, and with what degree of durability.

CONCLUSION

This systematic review synthesized 30 empirical studies published between 2020 and 2025 to examine the development trends, learning effectiveness, and research gaps pertaining to interactive digital comics in Indonesian language learning for lower-grade elementary school students. Three principal conclusions emerge from the evidence. First, research output on this topic has escalated significantly since 2021, driven by post-pandemic digital adoption, with Android and web-based platforms dominating deployment. Bibliometric analysis confirms a decisive shift in the field's theoretical orientation, from broad pedagogical discourse toward multimodal and multiliteracy frameworks, reflecting an institutional recognition of interactive digital comics as substantive literacy tools rather than supplementary aids. Second, interactive digital comics demonstrably improve reading comprehension, vocabulary mastery, and learning motivation, with 83.3% of included studies reporting positive outcomes relative to conventional instruction. These effects are theoretically explicable through Dual Coding Theory and the Cognitive Theory of Multimedia Learning, both of which predict enhanced encoding and retention when verbal and visual information channels are simultaneously and coherently engaged. Third, implementation remains constrained by uneven digital infrastructure, variable teacher digital competency, and the high interdisciplinary demands of quality media development, structural barriers that temper the scalability of the observed benefits. This review contributes to the field by providing the first systematic, PRISMA-guided synthesis specifically focused on interactive, rather than static, digital comics in Indonesian language instruction at the lower primary level, thereby establishing a more precise evidence base than prior general reviews of digital learning media. Future research should prioritize randomized controlled trials with adequate follow-up periods to assess durability of effects, pursue large-scale studies spanning diverse geographic and socioeconomic contexts, and explore the pedagogical value of integrating local cultural content into interactive digital comic design. Addressing these gaps is essential for translating the demonstrated potential of this medium into equitable, policy-supported improvements in Indonesia's early literacy outcomes.

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